

PYTORCH DDP SCALING BENCHMARK

B200 and RTX PRO 6000 on AICR

Capacity planning and baseline performance

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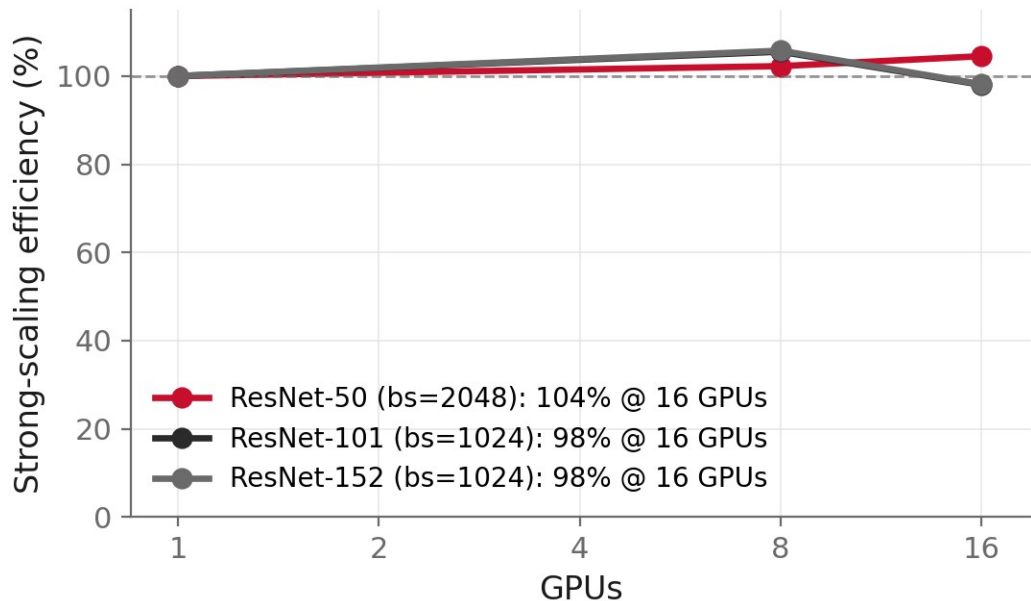
Test setup

Models	ResNet-50, ResNet-101, ResNet-152
Framework	PyTorch 2.10.0+cu130
CUDA	13.1 (sm_100 for B200, sm_120 for RTX PRO 6000 Blackwell)
Parallelism	DDP, NCCL backend, torchrun launcher
Precision	BF16, channels-last memory format
Per-GPU batch	128 on RTX PRO 6000 (96 GB), 256+ on B200 (192 GB)
Global batches	1024 and 2048 (or 512 / 1024 for ResNet-152 on 96 GB)
Scope	Strong and weak scaling, 1 to 16 GPUs (single + 2-node)

Warmup uses a probe-based step count agreed across ranks via `all_reduce MAX`. Avoids the wall-clock desync that caused an earlier DDP deadlock.

RESULTS

RTX PRO 6000: strong scaling



Key finding

Near 100% at moderate-to-large batches.

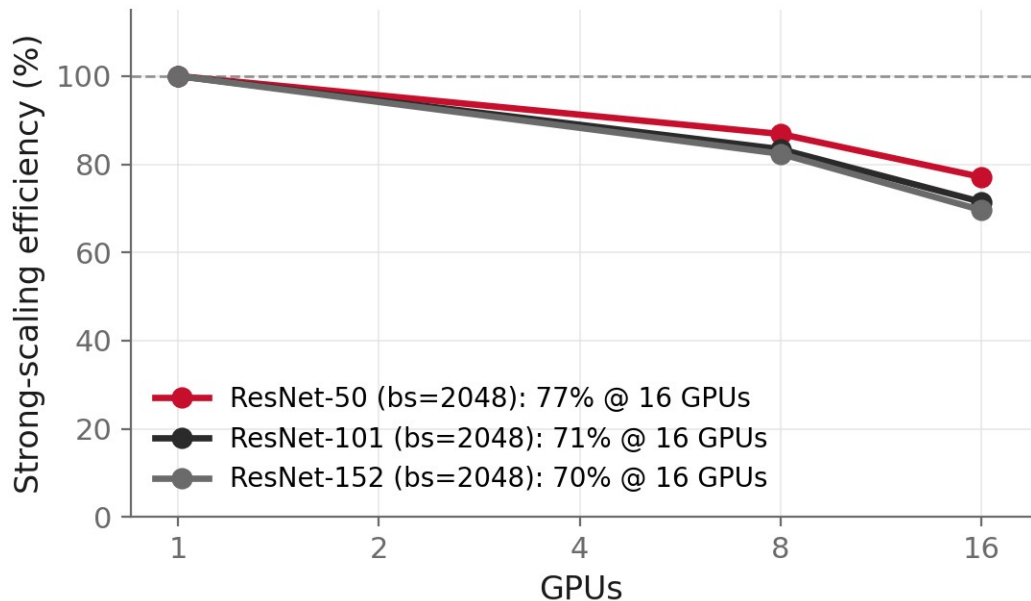
ResNet-50: 104% at bs=2048 (small superlinear, likely cache).

ResNet-101: 98% at bs=1024.

ResNet-152: 98% at bs=1024.

RESULTS

B200: strong scaling



Key finding

70 to 77% at 16 GPUs.

ResNet-50: 77% at bs=2048.

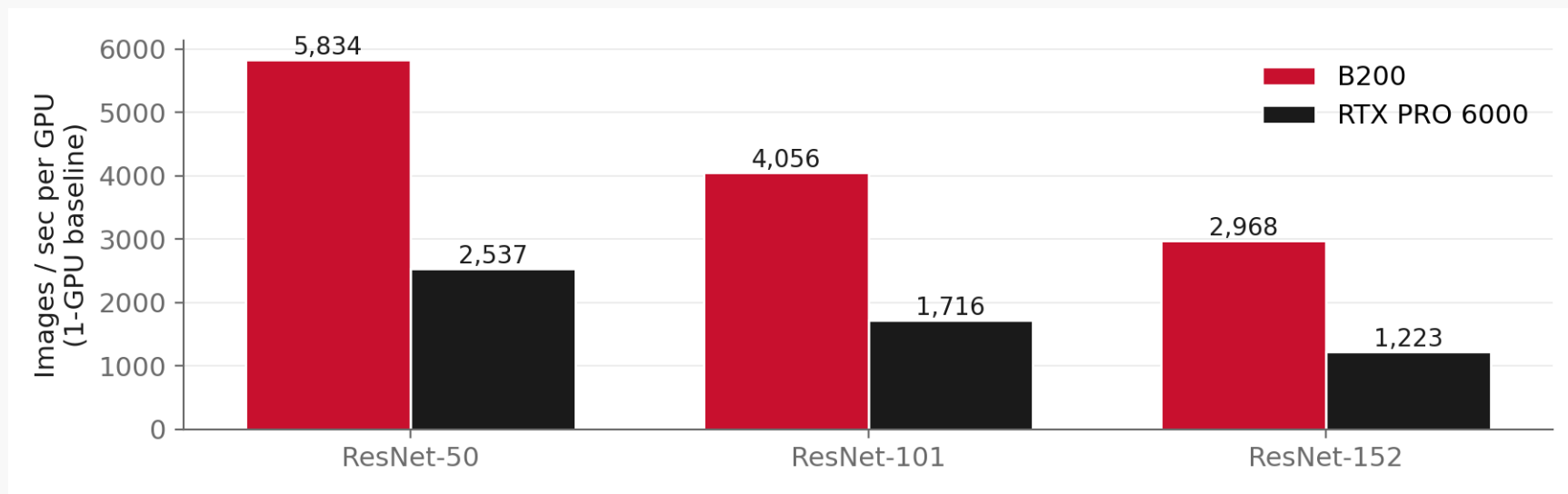
ResNet-101: 71% at bs=2048.

ResNet-152: 70% at bs=2048.

Faster compute = larger relative comm cost.

COMPARISON

Per-GPU throughput by GPU type



B200

Highest absolute throughput per GPU. Best for jobs that fit on a small number of GPUs or are compute-bound.

RTX PRO 6000

Lower per-GPU throughput, scales near-linearly at moderate batches. Best for strong-scaling-sensitive workloads.

Issues found and resolved

DDP warmup deadlock

What happened: Per-rank wall-clock timing caused ranks to disagree on step count, leading to a hang at the first collective.

Resolution: Replaced with probe-based step count, agreed across ranks via all_reduce MAX. Fix in the benchmark repo.

Node a0016 CUDA init failures

What happened: Persistent CUDA initialization failures on a single B200 node, reproducible across users.

Resolution: Filed support ticket. Resolved by node reboot. Returned to production.

B200 thermal observations

What happened: Thermal variation on some B200 nodes during validation.

Resolution: Reported. Published numbers verified unchanged at matched configurations after remediation.

Conclusions

Results are within expected ranges. AICR passed all the tests I tried and performs as expected.

- RTX PRO 6000 delivers near-linear strong scaling for ResNet-50 at all tested batches, and 98% for the deeper variants at bs=1024.
- B200 delivers the highest absolute throughput per GPU; strong-scaling efficiency tapers at the multi-node boundary as expected.
- Weak scaling is healthy on both card types from 1 to 16 GPUs. GPU activity stays above 90% across all configurations.
- Benchmark harness, deadlock fix, and analysis scripts are in the team repo for reuse on future driver or PyTorch updates.

User guidance and resources

User guidance

- Use the largest feasible global batch on B200.
- Choose RTX partitions when strong-scaling efficiency matters at adequate batches.
- Choose B200 partitions when raw FLOPs density or large per-GPU memory matters.

Resources

Benchmark code, sbatch templates, results

https://github.com/mghpcc/aicr-benchmarks/tree/main/Benchmark_WG/pytorch-training

Contact

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Questions?